

VALIDATION REPORT:
PHOTON ATTENUATION SIMULATOR
(V1.02)

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1. EXECUTIVE SUMMARY

This report documents the validation of the Shielding Attenuation Simulator (v1.02). The software is designed to calculate photon transmission and uncollided flux through multi-layer shielding geometries using the Beer-Lambert Law and the Inverse-Square Law. Validation was performed against analytical benchmarks, geometric scaling tests, and robustness checks for edge-case inputs. All validation tests performed in this report indicate that the implemented narrow-beam attenuation, interpolation, and inverse-square flux calculations operate within expected physical and mathematical bounds.

2. METHODOLOGY AND PHYSICS MODELS

2.1 Attenuation Model

The primary calculation for shielding effectiveness utilizes the Beer-Lambert Law. The simulator calculates the transmission (T) of a mono-energetic photon beam through N layers of material:

$$T = \exp\left(-\sum_{i=1}^N \mu_i x_i\right)$$

where:

- μ_i is the linear attenuation coefficient of the i-th material (cm^{-1}).
- x_i is the thickness of the i-th layer (cm).

2.2 Material Data and Interpolation

Attenuation coefficients are derived from NIST XCOM mass attenuation data (μ/ρ). For energies between tabulated data points, the simulator employs log-log interpolation to account for the power-law dependence of attenuation coefficients on energy.

$$\ln(\mu) = \ln(\mu_1) + \frac{\ln(E) - \ln(E_1)}{\ln(E_2) - \ln(E_1)} \cdot (\ln(\mu_2) - \ln(\mu_1))$$

2.3 Geometric Flux Calculation

The uncollided detector flux (ϕ) is calculated by combining the shield transmission with geometric spreading from a point source:

$$\phi = \frac{S \cdot T}{4\pi r^2}$$

where:

- S is the source strength in photons / s
- r is the distance from the source to the detector (cm).
- T is the dimensionless shield transmission.

3. TEST CASE RESULTS

3.1 Analytical Benchmarks

Test Case	Input Parameters	Expected Result	Actual Result	Status
1	1 Layer Pb, 5 cm, 0.6617 MeV,	0.00183626 Transmission	0.0018362592 Transmission	PASS
2	1 Layer Al, 10 cm, 0.6617	0.133400 Transmission	0.13340005 Transmission	PASS
3	1 Layer Pb (5 cm) -> 1 Layer Al (10cm), 0.6617 MeV	0.000244957 Transmission	0.00024495705 Transmission	PASS
4	1 Layer Pb, 5 cm, 1.0 MeV (exact NIST tabulated value)	$\mu/\rho = 0.07102$	$\mu/\rho = 0.07102$	PASS
5	1 Layer Pb, 5 cm, 0.002484 MeV (duplicate table edge)	$\mu/\rho = 1396$	$\mu/\rho = 1396$	PASS
6	1 Layer Pb, 0 cm, 0.6617 MeV	1.0 Transmission	1.0 Transmission	PASS
7	Invalid Distance	ValueError	ValueError	PASS
8	Pb HVL / TVL: 1 Layer Pb, .550 cm and 1.827 cm, 0.6617 MeV	HVL: 0.5 Transmission TVL: 0.1 Transmission	HVL: 0.50007223 Transmission TVL: 0.10005562 Transmission	PASS

Note: The Half-Value Layer (HVL) thickness for Lead at 0.6617 MeV was calculated as 0.550 cm, confirming the accuracy of the NIST data interpolation and the $\ln(2)/\mu$ relationship. The Tenth-Value Layer (TVL) thickness for Lead at 0.6617 MeV was calculated as 1.827 cm, confirming the accuracy of the NIST data interpolation and the $\ln(10)/\mu$ relationship.

3.2 Geometric Scaling (Inverse Square Law)

This test verified the $1/r^2$ relationship by doubling the detector distance while maintaining zero shielding.

- Flux at 10 cm: $2.944366 \cdot 10^7$ photons / cm² / s
- Flux at 20 cm: $7.360916 \cdot 10^6$ photons / cm² / s
- Observed Ratio: 3.9999995
- Theoretical Ratio: 4.0
- Status: PASS

3.3 Robustness and Error Handling

The software was subjected to invalid inputs to verify exception handling.

- Energy Out of Bounds: System correctly identified energy out of bounds and handled the exception via ValueError for 0.0005 MeV identifying it as outside the NIST range of 0.001-10.0 MeV.
- Data Integrity: Successfully caught ValueError when negative attenuation coefficients were manually injected into the material library.
- Duplicate Data Points: Successfully handled duplicate energy entries in NIST tables (e.g., at 0.002586 MeV for Pb) without division by zero errors. Automatically selected the post-edge duplicate value according to the simulator's duplicate-energy policy.
- List Mismatch: Correctly identified cases where material library lists had different lengths and threw ValueError.
- Negative Source Strength: Successfully raised ValueError when given negative source strength.
- Negative Transmission: Successfully raised ValueError when impossible configuration resulted in negative transmission value.
- Negative Distance: Successfully raised ValueError when given negative detector to source distance value.
- Negative Linear Attenuation Coefficient: Successfully raised ValueError when impossible configuration resulted in negative linear attenuation coefficient.
- Negative Photon Energy: Successfully raised ValueError when given negative photon energy.
- Negative or Incorrect Material Values: Successfully raised ValueError for any case of impossible material properties such as negative density or negative μ/ρ values.

4. ASSUMPTIONS AND LIMITATIONS

1. Uncollided Flux Only: The current version calculates uncollided flux. Buildup factors to account for scattered radiation contributions are not yet implemented.
2. Narrow Beam Geometry: Calculations assume narrow-beam conditions where any interaction removes the photon from the beam.
3. Point Source: The source is modeled as an isotropic point source.

5. CONCLUSION

The Shielding Attenuation Simulator (v1.02) is validated for use in modeling primary photon beam attenuation. The implementation of the Beer-Lambert Law, NIST data interpolation, and geometric spreading has demonstrated accuracy across all tested scenarios. The software is now ready for the next step: integration of ORNL-standardized buildup factors to improve accuracy in broad-beam shielding scenarios.

Bibliography

- [1] Angstrom Sciences, “Density of elements chart – Angstrom Sciences elements density table,” Angstrom Sciences, <https://angstromsciences.com/density-elements-chart> (accessed Mar. 26, 2026).
- [2] J. Colledge, M. Wilczek, and C. Raymond, “Inverse square law | radiology reference article,” Radiopaedia, <https://radiopaedia.org/articles/inverse-square-law?lang=us> (accessed Mar. 26, 2026).
- [3] “Shielding Radiation Alphas, Betas, Gammas and Neutrons,” U.S.NRC, <https://www.nrc.gov/docs/ML1122/ML11229A721.pdf> (accessed Mar. 26, 2026).
- [4] M. J. Berger et al., “XCOM: Photon Cross Sections Database,” NIST, <https://dx.doi.org/10.18434/T48G6X> (accessed Mar. 26, 2026).